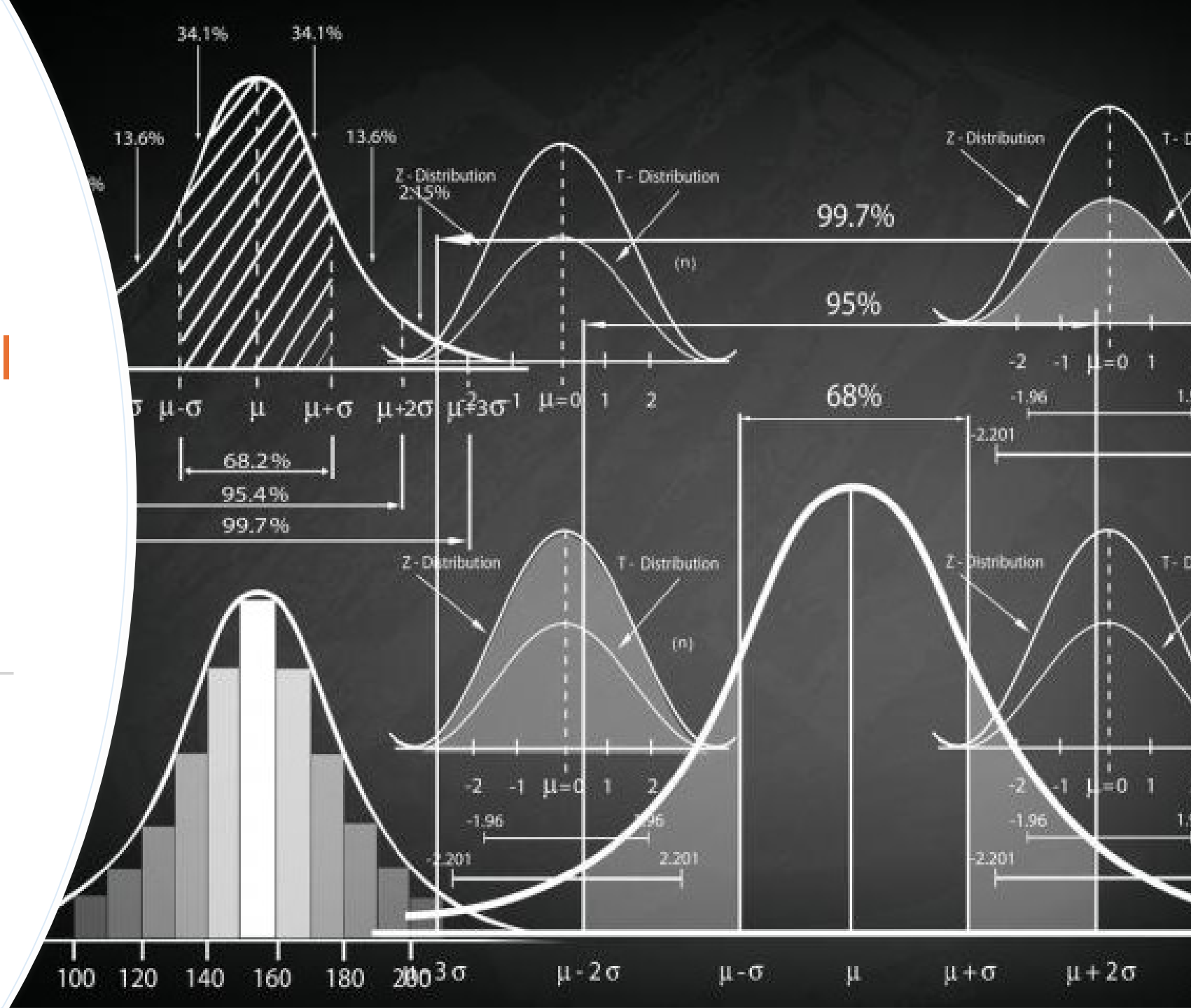
Comparing Mutual Information in Finite Samples

Normal Wald and Expanded Welch–Satterthwaite Tests

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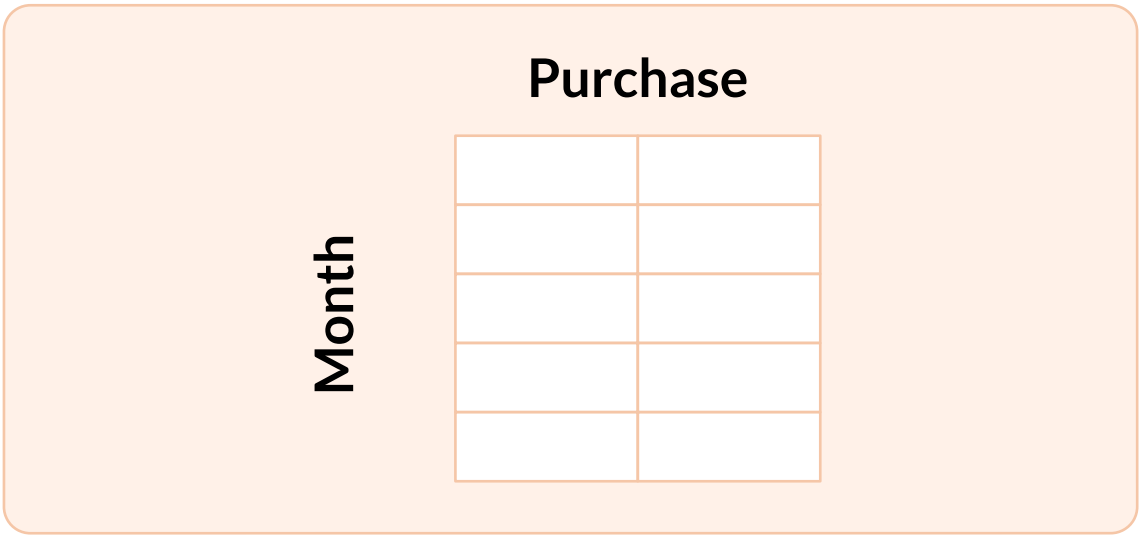


Comparing association across populations

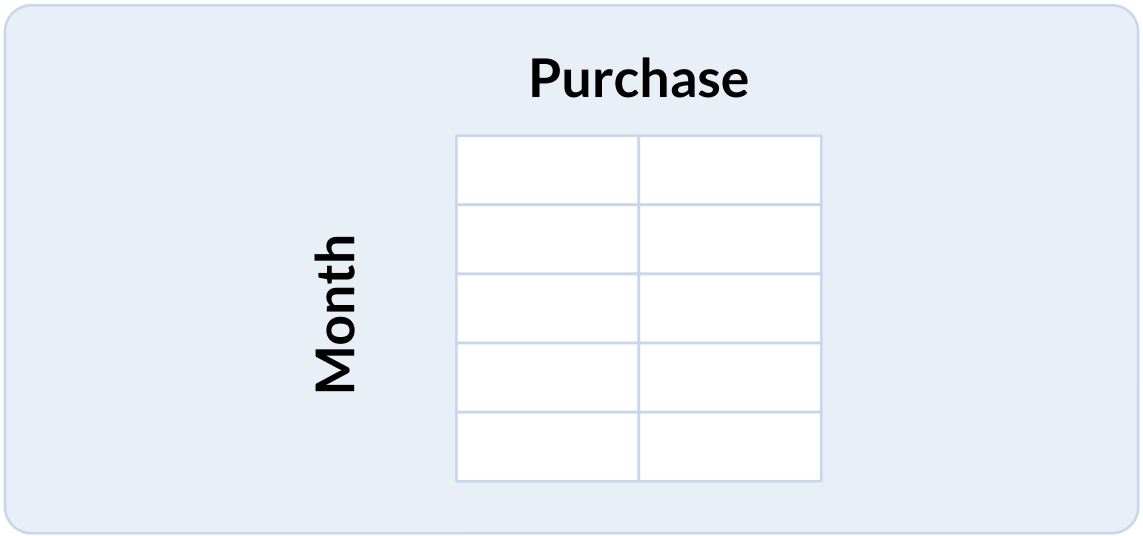
Mutual information: average reduction in uncertainty about one variable obtained by observing another

Month of visit and purchase outcome

P: new visitors



Q: returning visitors



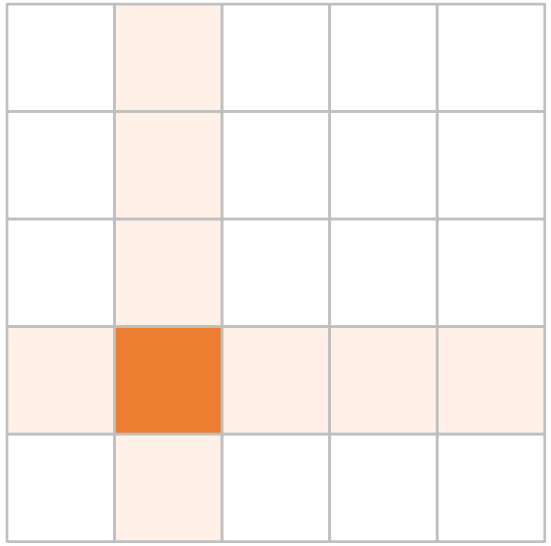
Does the month–purchase MI differ between these populations?

Bossomaier et al. (2016); data: Sakar and Kastro (2018). Thesis Sections 1.1, 2.1 and 6.4

Sampling uncertainty in the MI comparison

$$\Delta = I(P) - I(Q), \qquad H_0 : \Delta = 0$$

- Rare category combinations may be missed or observed only once
- Errors can affect both the estimated MI difference and its standard error
- Under equal population MI, the intended rejection rate is **5%**



Schematic: a rare cell, its row and its column

Existing foundations and the MI-specific contribution

Part of the method	Role in this project
MI bias and sampling variance	Estimate the difference and its uncertainty
Welch–Satterthwaite moment matching	Choose a reference from variance-estimator variability
MI-specific variance sensitivity, derived here	Track changes in cell probabilities, margins and pointwise MI

Moddemeijer (1989); Paninski (2003); Satterthwaite (1946); Welch (1947); Hutcheson (1970); Pan and Wall (2002). Thesis Sections 2.2–2.5

Normal Wald and Expanded Welch

Same estimates and statistic, larger rejection cutoff

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}$$

BC: leading bias correction

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}$$

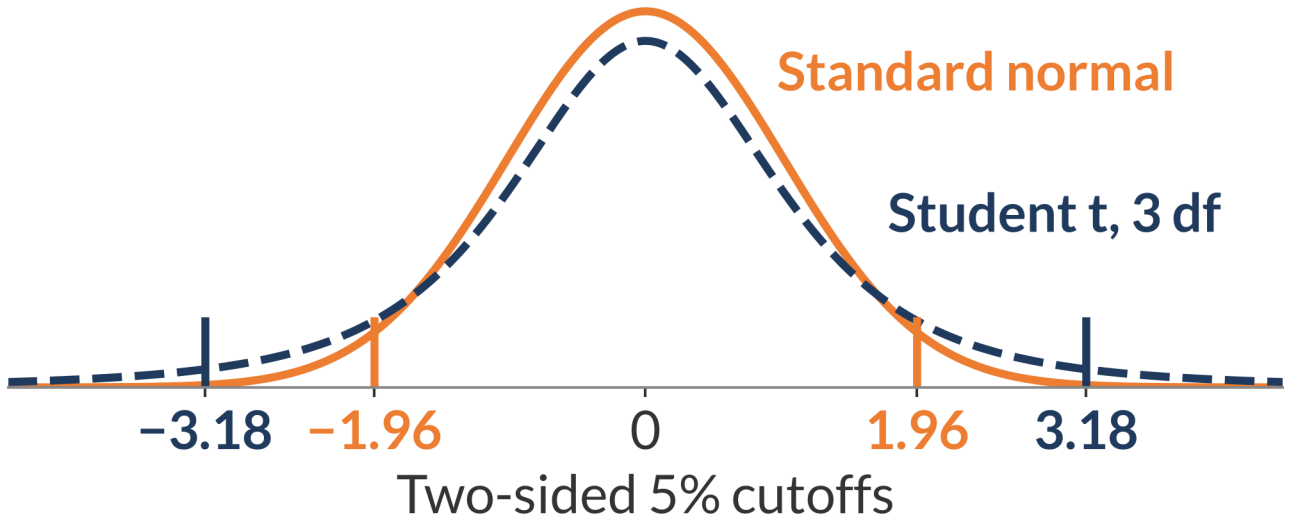
Normal Wald

Standard normal reference

Expanded Welch

Student t reference with estimated degrees of freedom

Illustration



Illustrative Student t distribution with 3 degrees of freedom. Thesis Sections 3.2–3.3 and 4.1–4.3

Deriving the degrees of freedom

1 **Sensitivity of the pointwise-MI variance**

P_ε moves a small amount of probability towards cell $(x,\ y)$

$$g_P(x, y) = \frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\Bigg|_{\varepsilon = 0}$$

Track joint, row and column changes together

2 **Sampling variation of its estimate**

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}, \qquad \tau^2(P) = \mathrm{Var}_P\{g_P(X, Y)\}$$

3 **Component degrees of freedom**

$$\hat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\hat{\tau}^2(P)}$$

Repeat for Q and combine the components using **Welch–Satterthwaite moment matching**

Satterthwaite (1946); Welch (1947). Thesis Sections 4.1–4.3

Evaluation from known populations

Research questions

RQ1: Approximation accuracy

Mean and sampling variance of $\hat{V}$, including near independence

RQ2: False positives

Equal MI: 0.005, 0.02 and 0.30 nats

RQ3: Power

MI in P = 0.02; gaps of 0.005, 0.01 and 0.02 nats

Shared settings

- Four table sizes: 2×2, 4×4, 8×8, 16×16
- Uniform, same-skew and different-skew margins
- Ten sample sizes, 10 to 50,000 observations per population
- 100,000 independent table pairs per setting

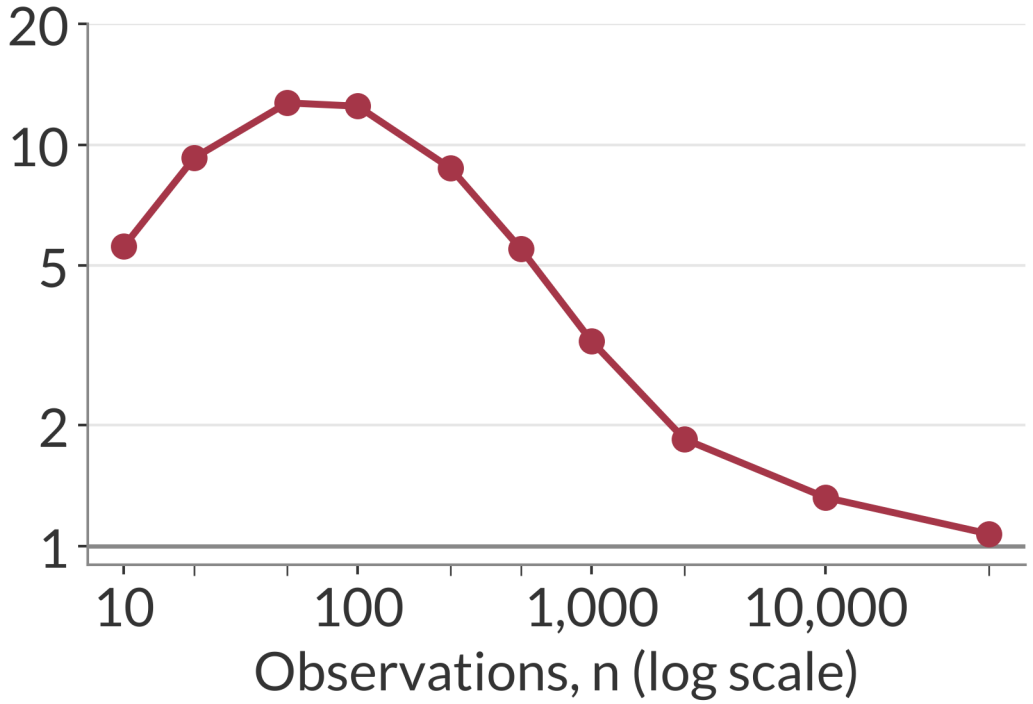
Both methods use each same pair of sampled tables. Invalid outputs count as non-rejections; validity is reported.

Morris et al. (2019). Thesis Chapter 5, Tables 5.1–5.3; construction and validity rules in Appendix B

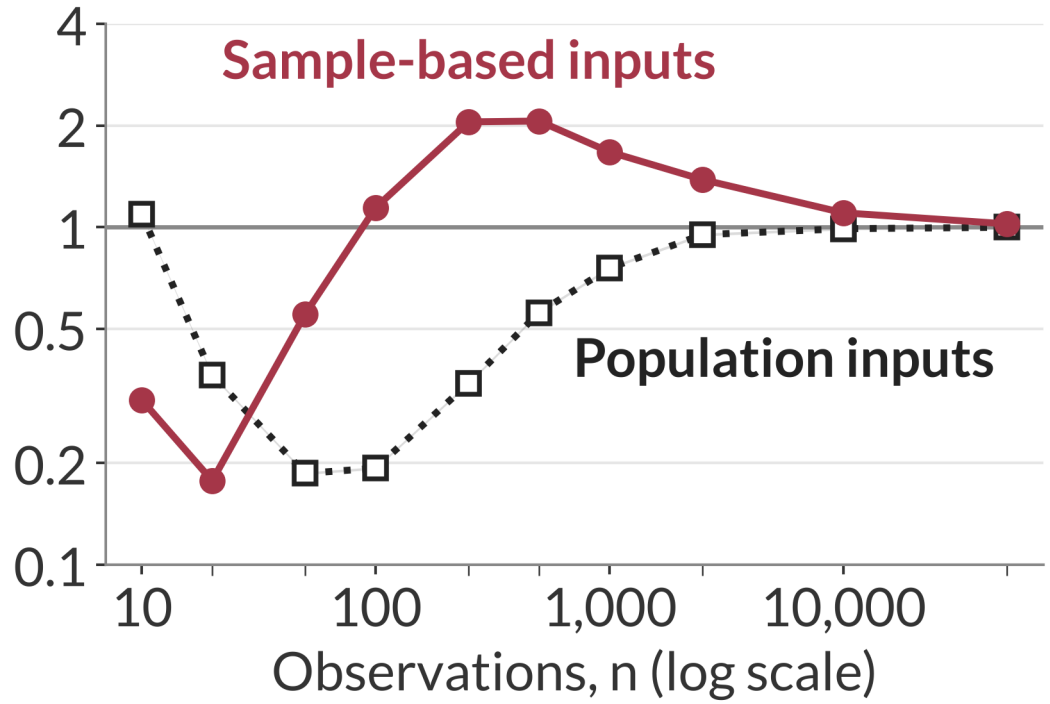
Sparse samples distort the variance inputs

Different-skew P, 16×16, population MI 0.02 nats

Mean estimate / population variance



Variance prediction / observed sampling variance

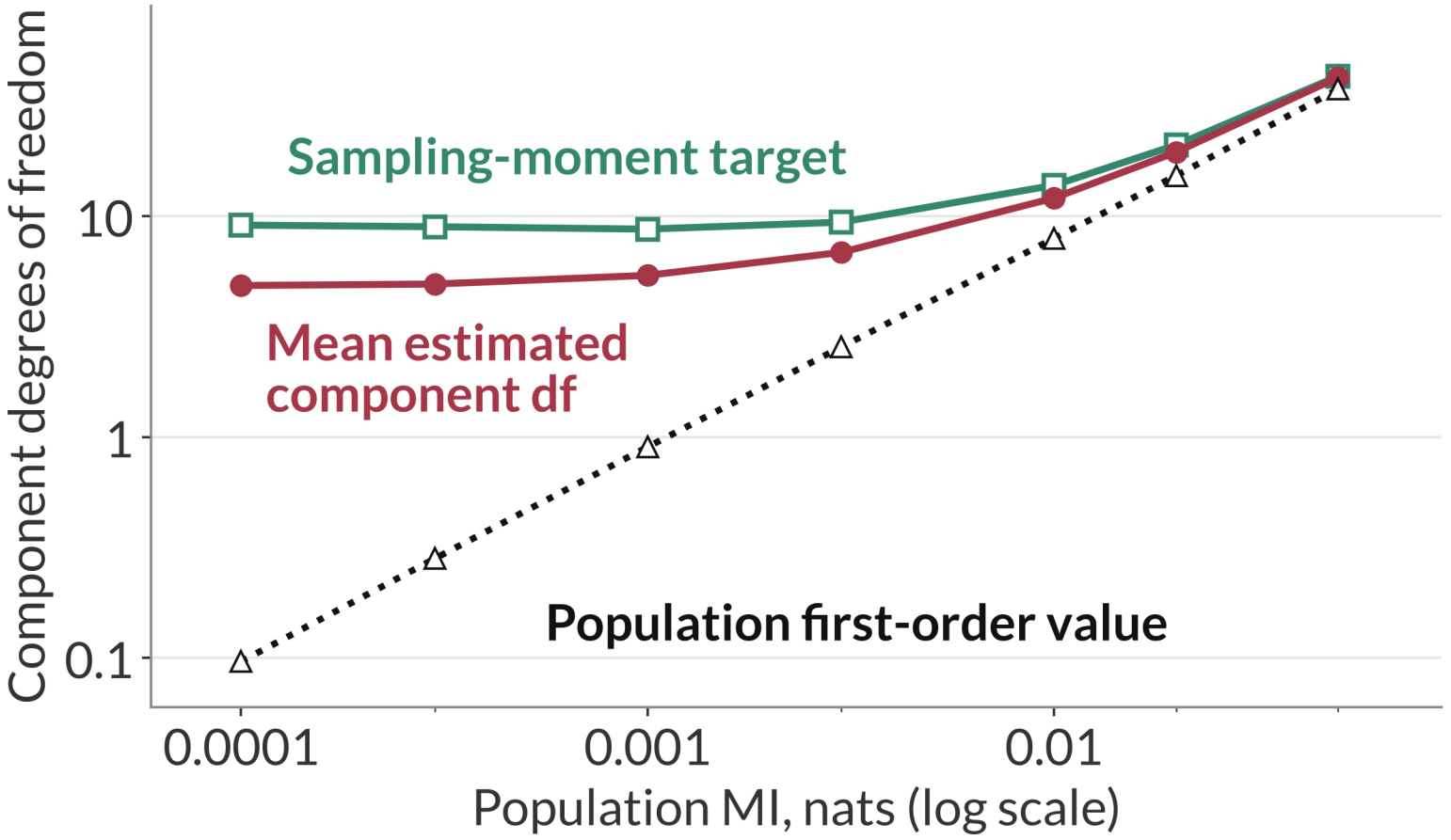


One indicates agreement. Errors affect both parts of the degrees-of-freedom calculation.

Illustrative slice of the RQ1 study. Thesis Section 6.1.1, Figure 6.1. Bands: 95% Monte Carlo intervals

Sampling noise matters near independence

Different-skew P, 4×4, n = 1,000 per population



Near independence

At MI 0.0001:

- mean estimate about 5
- first-order value about 0.1

Quadratic sampling variation creates apparent association

Thesis Section 6.1.3, Figure 6.3; Appendix A.2. Bands: 95% Monte Carlo intervals

The benefit depends on Wald's false-positive rate

Equal population MI = 0.02 nats; target false-positive rate = 5%

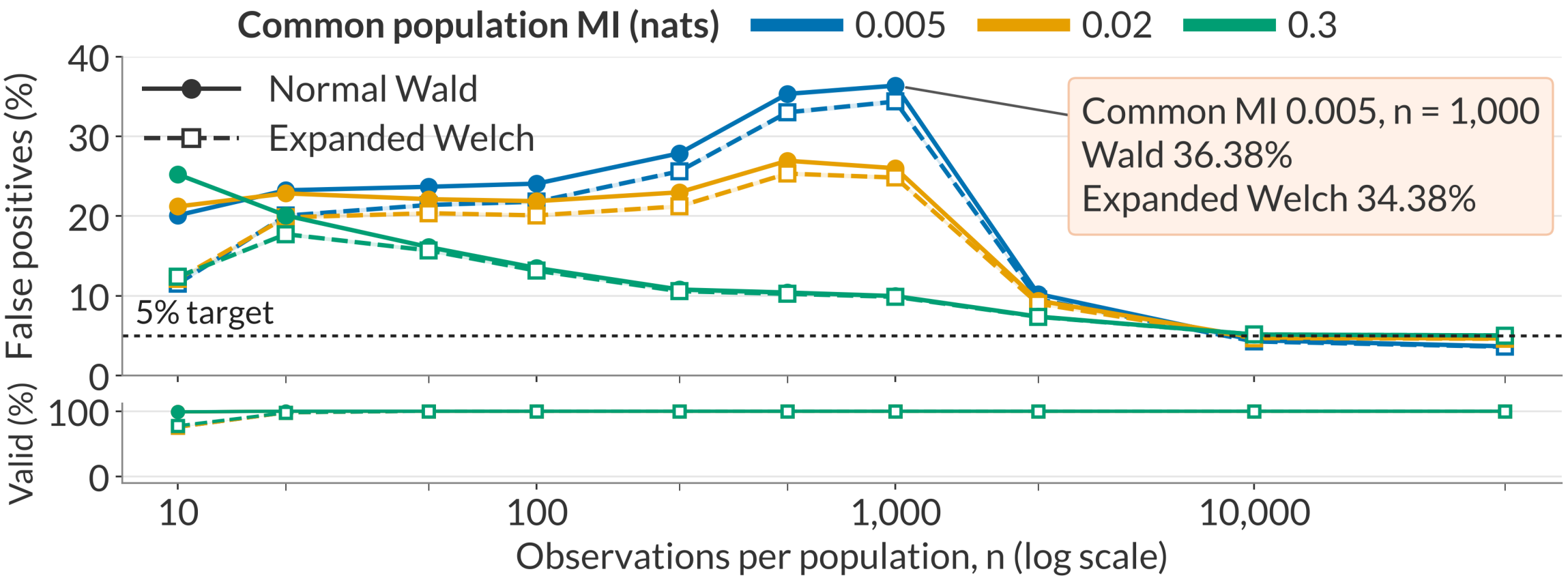
Illustrative setting	Normal Wald	Expanded Welch	Interpretation
Uniform 4×4 n = 20	8.35%	**4.90%**	Closer to 5%
Same skew 16×16 n = 1,000	0.60%	0.53%	Further below 5%
Different skew 16×16 n = 1,000	26.01%	24.84%	Still far above 5%

Both methods are valid in all 100,000 repetitions in each example.

Illustrative diagnostic rows from the balanced null grid. Thesis Section 6.2.4, Table 6.1

False-positive rates across common MI values

16×16, different-skew margins; equal MI in P and Q

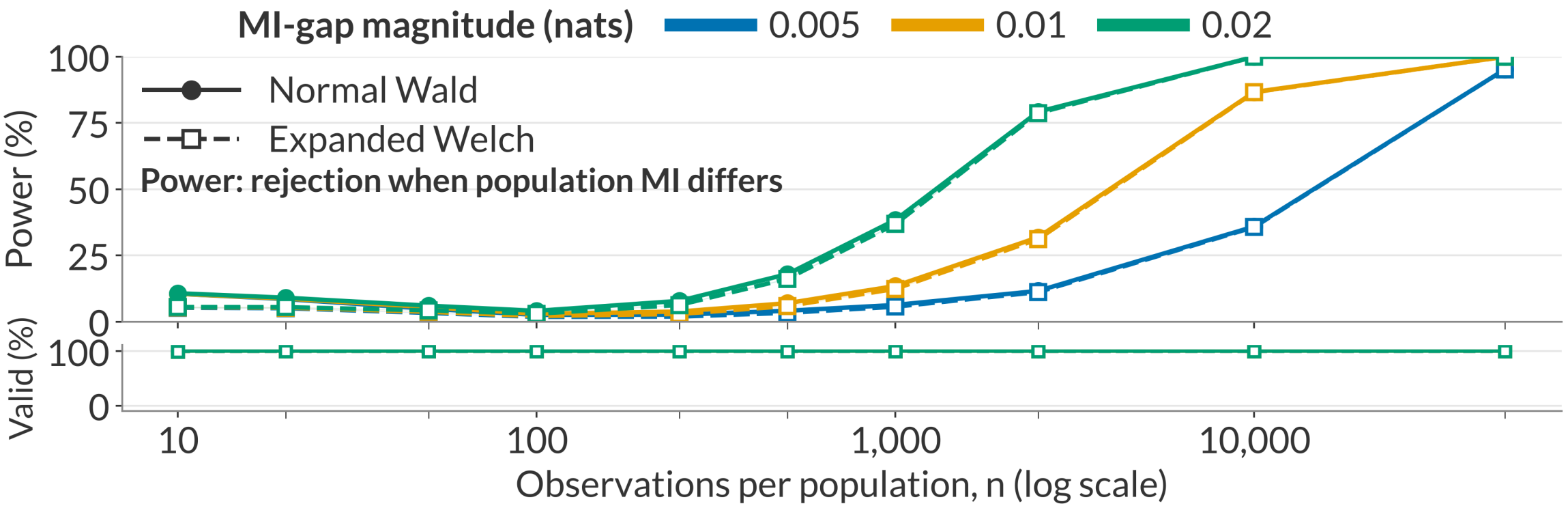


Both methods are valid in every repetition at this point

Thesis Section 6.2.3, Figure 6.6. Bands: pointwise 95% Monte Carlo intervals

Power across three MI differences

Uniform 4×4; MI in P = 0.02 nats; usual 0.05 p-value threshold for both methods

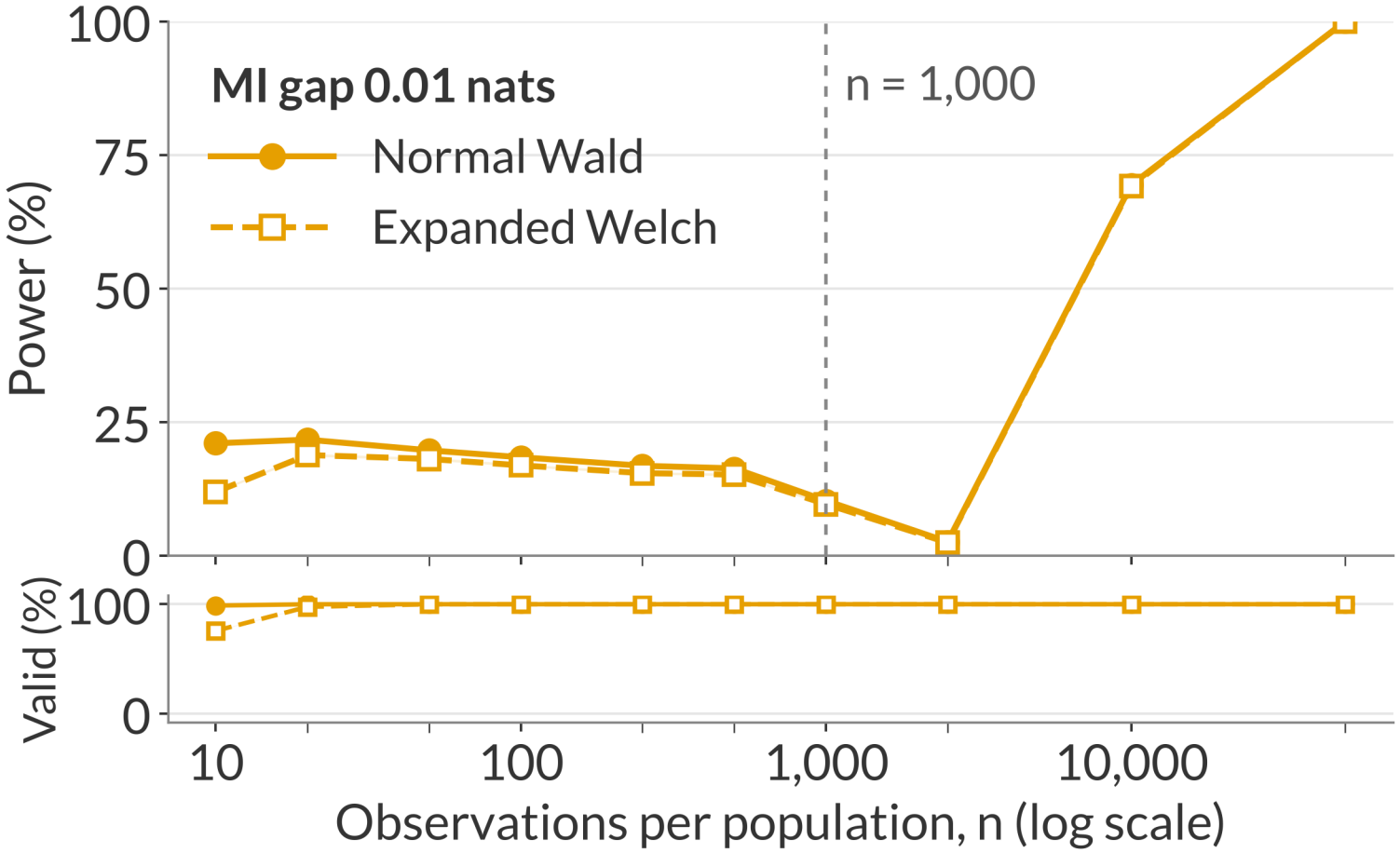


With sufficient observations, larger MI gaps are easier to detect. **Expanded Welch has a lower or equal rejection rate.**

Thesis Section 6.3.1, Figure 6.7. Bands: pointwise 95% Monte Carlo intervals

Bias can reverse the estimated direction

Different-skew 16×16; MI 0.02 versus 0.03 nats



At n = 1,000 per population

True difference

−0.0100 nats

Mean estimated difference

+0.0207 nats

Among Expanded Welch rejections, over 99.9% have the opposite estimated direction

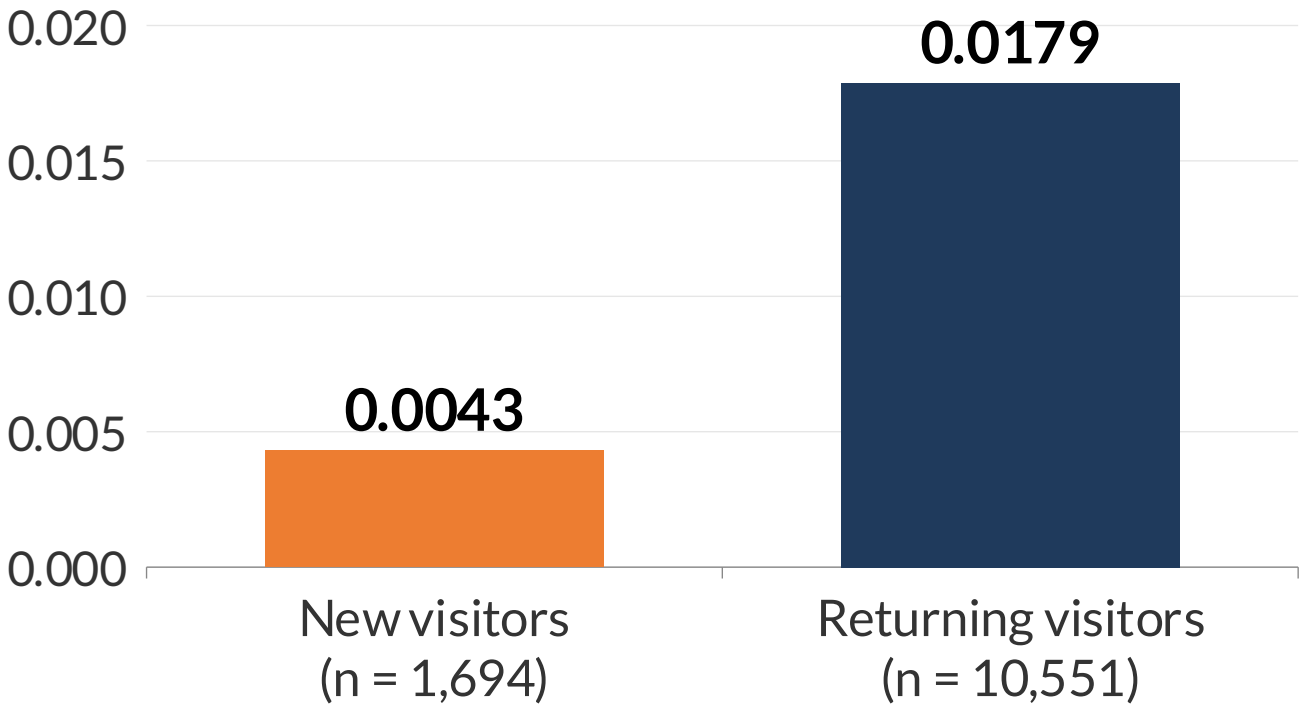
Unequal residual biases can outweigh the true gap

9,659 of 9,662 Expanded Welch rejections at n = 1,000 have a positive estimated difference. Thesis Section 6.3.3, Figure 6.9

An application to online shopping

Does month–purchase MI differ between new and returning visitors?

Corrected MI estimate (nats) point estimates



Test result

Estimated difference, new minus returning

−0.0136 nats

Both tests reject after **Holm adjustment** across six comparisons

Expanded Welch adjusted
p = **0.00149**

Observed-data illustration; population MI is unknown

Data: Sakar and Kastro (2018), UCI Machine Learning Repository. Holm (1979). Thesis Section 6.4, Tables 6.2–6.3

Limitations and future work

Present limit	Next investigation
First-order approximation near independence	Retain the additional quadratic sampling variation
Bias and errors in the shared standard error	Evaluate improved estimates one at a time
One population construction; equal simulation sample sizes	Test other association patterns and unequal sample sizes

Contributions and conclusions

Contributions

1

Derived MI-specific **variance sensitivity** and degrees of freedom

2

Improved **some** excessive false-positive rates, with a reduction in power

3

Sparse-table failures identify **estimation accuracy** as a priority

Thank you

Variance sensitivity

Perturbation towards one cell: $P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}$

Changing probability weights

$$g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P)$$

Changing pointwise-MI values and margins

$$+ 2[\ell_P(x,y) - \mathbb{E}_P\{\ell_P(X,Y) \mid X = x\}$$

$$- \mathbb{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)]$$

Pointwise MI: $\ell_P(i,j) = \log\{p_{ij}/(p_{i+}p_{+j})\}$

Sensitivity variance: $\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}$

Exact for positive population tables; its use for sampling errors is first order. Thesis Chapter 4

Moment matching and the combined reference

Component degrees of freedom

$$\nu_{P,\,\mathrm{mom}} = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}, \qquad \widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}$$

Combined degrees of freedom

$$\widehat{\nu} = \frac{\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\}^2}{\{\widehat{V}(P)/n_P\}^2/\widehat{\nu}_P + \{\widehat{V}(Q)/n_Q\}^2/\widehat{\nu}_Q}$$

Moment matching does not establish the normality or independence needed for an exact Student t distribution.

Satterthwaite (1946). Thesis Sections 4.1 and 4.3

Population construction and parameter grid

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}\{i=j\}, \qquad 0 \le \lambda < 1$$

Margin vector: $m_k(\rho) = (\rho,\ (1-\rho)/(k-1),\ \ldots,\ (1-\rho)/(k-1))$

Table size, k	2, 4, 8, 16
Sample size, n	10, 20, 50, 100, 250, 500, 1,000, 2,500, 10,000, 50,000
Uniform	Both margins uniform in both populations
Same skew	P and Q each use $m_k(0.8)$
Different skew	P uses $m_k(0.7)$; Q uses $m_k(0.8)$
Null	Common MI 0.005, 0.02, 0.30 nats
Power	P MI 0.02; Q MI 0.025, 0.03, 0.04 nats

Thesis Section 5.1, Tables 5.1–5.2; Appendix B.1–B.2

Validity and rejection denominators

Normal Wald is valid with

- a finite difference and finite positive standard error
- $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$
- a finite p-value

Expanded Welch is valid with

- the Wald requirements
- positive finite component and combined degrees of freedom
- a finite p-value

$$\text{rejection rate} = \frac{\text{valid rejections}}{\text{all repetitions}}$$

Validity is reported separately.

An invalid calculation is not evidence that the null is true.

Zero and undefined component degrees of freedom are different cases; both are invalid.

Thesis Appendix B.3–B.5; validity strips in Figures 6.4–6.9

The near-independence noise calculation

Setting: 4×4, n = 1,000, MI 0.0001; $d = 9$ **and** $\kappa = 2nI(P) \approx 0.2$

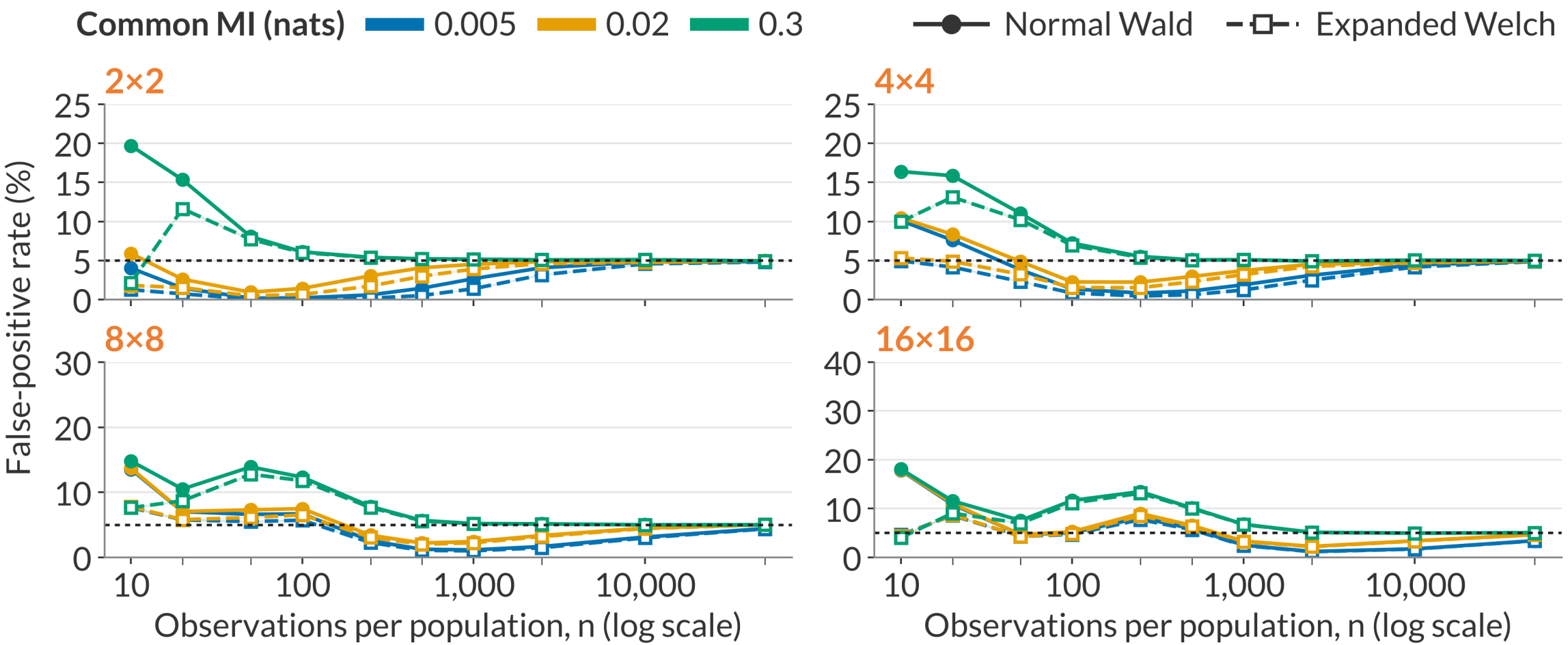
$$G = 2n\hat{I}(P) \;\dot{\sim}\; \chi^2_d(\kappa), \qquad \hat{\nu}_P \approx G/2$$

Quantity	Local prediction	Observed (P)
Mean estimated component df	4.60	4.86
Sampling-moment target	9.00	9.12
Mean sample-based variance prediction / observed sampling variance	1.96	1.97

Local approximations under fixed dimensions and positive limiting cell probabilities. Thesis Appendix A.2, Section 6.1.3

Null results with uniform margins

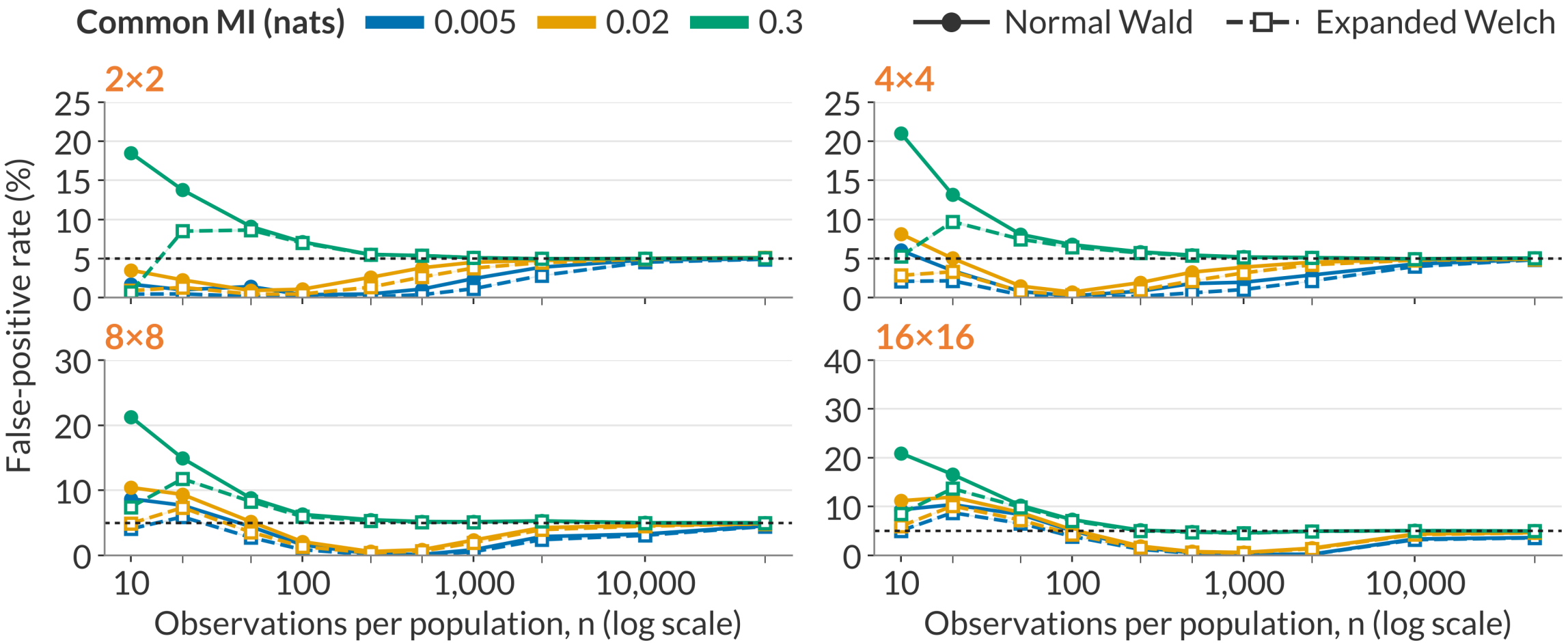
Equal MI in P and Q; uniform margins in both populations



Rejection rates use all 100,000 repetitions; invalid outputs count as non-rejections (validity: B4). Thesis Figure 6.4

Null results with same-skew margins

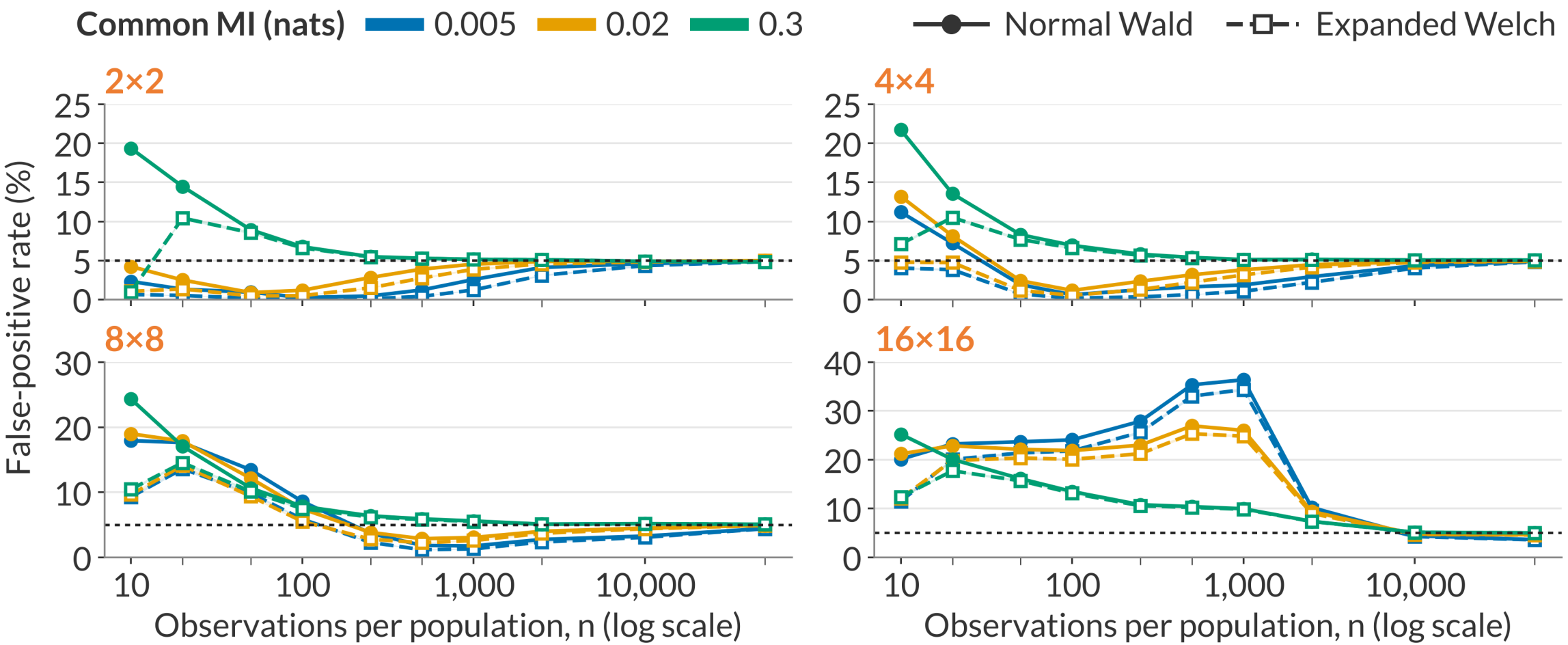
Equal MI; same-skew margins $m_k(0.8)$ in P and Q



Rejection rates use all 100,000 repetitions; invalid outputs count as non-rejections (validity: B4). Thesis Figure 6.5

Null results with different-skew margins

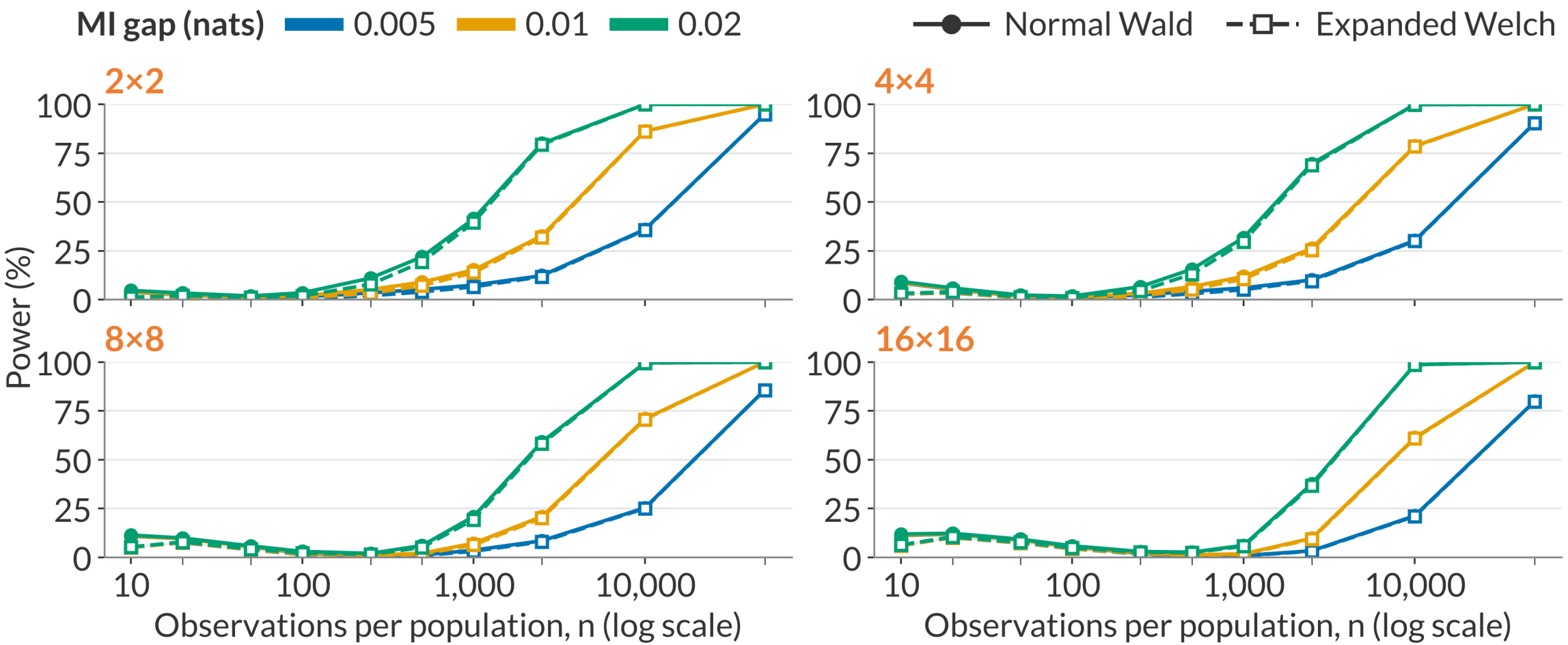
Equal MI; $m_k(0.7)$ in P and $m_k(0.8)$ in Q



Rejection rates use all 100,000 repetitions; invalid outputs count as non-rejections (validity: B4). Thesis Figure 6.6

Power with same-skew margins

Same-skew margins $m_k(0.8)$ in P and Q; MI in P = 0.02 nats



Rejection rates use all 100,000 repetitions; invalid outputs count as non-rejections (validity: B4). Thesis Figure 6.8

Application details and the comparison family

Six exploratory comparisons; Holm adjustment applied separately to each method

Dataset	Predictor	Normal Wald	Expanded Welch
Adult	Occupation	1.184×10^{-44}	1.635×10^{-42}
Adult	Education	1.890×10^{-34}	4.930×10^{-33}
Online Shoppers	Month	4.714×10^{-5}	3.726×10^{-4}
Student Performance	Study time	0.1982	0.2339
Student Performance	Maternal education	0.8361	0.8468
Online Shoppers	Weekend	0.9554	0.9601

Unadjusted p-values. Holm-adjusted month p-values: Normal Wald **0.000189**; Expanded Welch **0.00149**

Becker and Kohavi (1996); Sakar and Kastro (2018); Cortez (2008); Holm (1979). Thesis Appendix B.7, Table B.1

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